

Massive Attack Mapping Methodology

Kyiv 02/06/2026 Russian Bombardment GIS Analysis

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Abstract

Since I first began geolocating conflict zones, I felt that the process of identifying strikes was being done in reverse. In real life terms, militaries select strategic objectives, attempt to hit those objectives, and collateral damage ensues. Journalists, watchdog organisations and investigators will geolocate the scattered results from the collateral damage, and attempt to piece the story together.

However, this common methodology is disconnected from how intelligence operatives who do the targeting of strikes think. As a result, the pattern of destruction reveals itself only at the end of an investigation rather than the locations being distilled from a general pattern or principle, which would be far more efficient.

Therefore, I propose the idea that reverse engineering military intelligence requires a basic understanding of military tactical operations, and therefore work from a target infrastructure mapping as they do. If International Humanitarian Law investigators are to efficiently analyse strikes, they should have an idea of high priority target zones, and initiate analysis from those vicinities, starting their investigations from a targeting theory whenever they can.

This Kyiv study is a demonstration of a proof-of-concept of one possible such implementation. This paper proposes a target-centric methodology that seeks to reconstruct the logic of a strike campaign before individual strike locations are fully verified. Rather than beginning with confirmed damage sites, the approach begins with indicators of attack activity and the distribution of strategically relevant infrastructure. The objective is to generate prioritised areas of interest that can subsequently be examined through traditional OSINT methods.

Introduction

This GIS hotspot analysis is a lead-generation methodology. It narrows a large urban battlespace into a manageable set of locations for imagery review, geolocation, and source verification.

The event I have analysed in this study is the Russian attack on Ukraine, 1–2 June 2026, focusing on Kyiv.

The model is not designed to identify exact strike locations. Rather, it identifies areas where significant thermal activity occurred and highlights critical infrastructure located within the spatial proximity of the satellite detection. The resulting dataset should be treated as a prioritization tool for further investigation rather than a list of impacted facilities. Its function is to narrow the search space (but not in an absolute sense).

The primary 375m zone marked in red reflects the nominal spatial uncertainty of VIIRS detections. A secondary zone extending to 1 km was examined to identify potentially affected infrastructure that may not have generated a thermal signature directly beneath the facility itself, such as blast damage, debris impacts, or nearby fires.

There are 3 deliverables in this study:

1. Mean cluster coordinates.
2. 2 .klm files that can be overlaid on Google Earth Pro: *Fire Radius Overlay.kml* and *Bomb Radius Overlay.kml*
3. 2 spreadsheets which show the operational relevance to the Russian bombing campaign: strategic, operational and background infrastructure of the damage zone. *Primary zone KML (0–375 m)* and *Secondary zone KML (375–1000 m)*.

Recency:

This project was completed four days after the event.

Data source:

- FIRMS VIIRS thermal anomaly detections.
- OSM infrastructure data.

Number of hotspots analyzed:

45

Number of infrastructure objects identified:

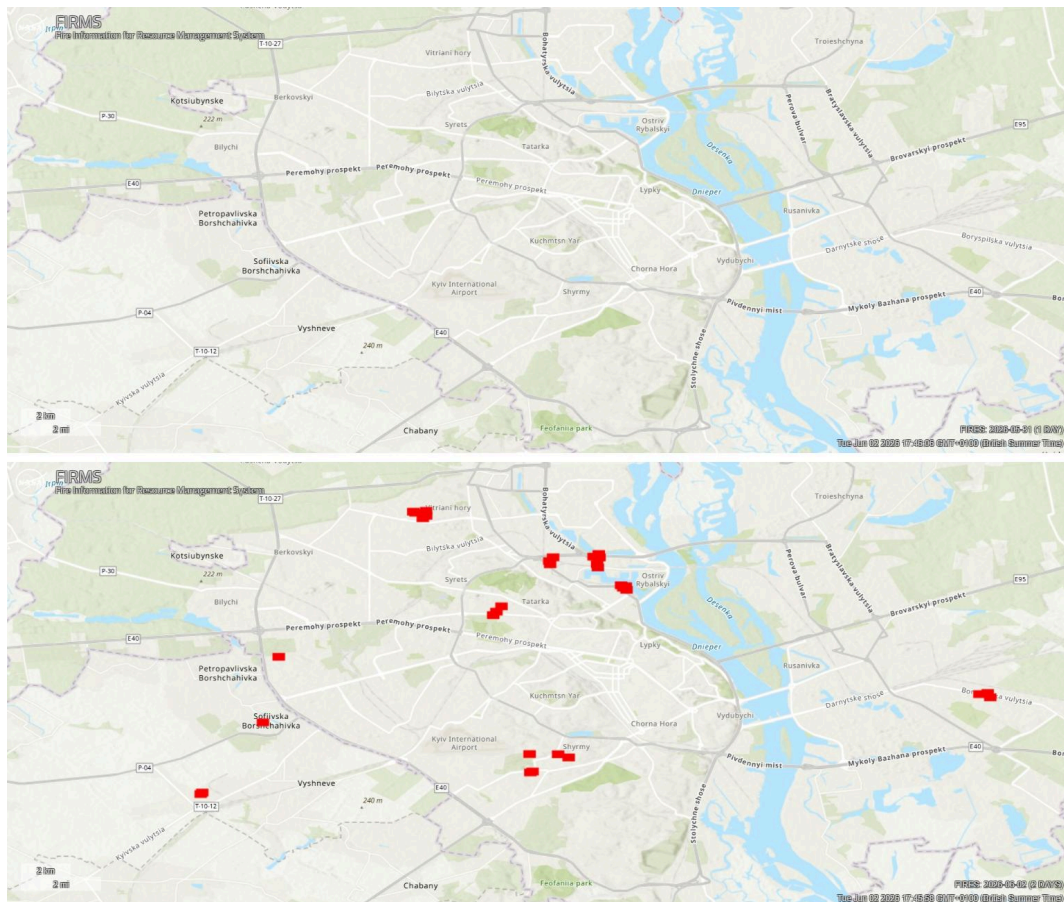
>200

Methodology

While monitoring the situation, I noticed that [fire breakouts](#) were very common around the city. I decided to check NASA's FIRMS website. The last time there were signs of heat anomaly in Kyiv was the 24th of May where there was another large-scale Russian bombing of the city. There were no detections since then until around midnight of 1st of June.

Given the facts, it was reasonable to infer that the fires meant:

- a - There was a bombing of the district or area of the clusters.
- b - There was either a high density of bombing or high infrastructure damage.



[Screen Captures from NASA FIRMS](#)

Such data leads to many assumptions and had inherent limitations, such as:

- The origin of a single spot could be from any building/infrastructure in the fire overlay.
- VIIRS hotspots represent thermal activity which can be caused by many reasons. They do not confirm impact points.
- Infrastructure proximity does not imply direct damage.
- Secondary-zone infrastructure should be considered, however debris damage is unpredictable and can easily be further away than 1km radius.
- There may be many locations of bombing that didn't create a fire signal.

Yet its advantages made it worthwhile to contextualise the given information.

I obtained the thermal anomaly detections for the 1–2 June attack by getting NASA FIRMS API. I merged detections from both days and both VIIRS NAA-20 and NAA-21 (URT+NRT) into a single dataset. I then remove duplicates.

API / area

Area

world

Date

2026-06-02

Source

VIIRS NOAA-20 (NRT+URT)

* empty if getting most recent data

Day Range

1

Map Key

3eb10b9840266c86121cd73

Display Results

Click or copy link generated below for your download script.

[/api/area/csv/3eb10b9840266c86121cd7399357789b/VIIRS_NOAA20_NRT/world/1/2026-06-02](#)

Mapkey transaction amount: 36

latitude,longitude,bright_t14,scan,track,acq_date,acq_time,satellite,instrument,confidence,version,bright_t15,frp,daynight

65.82642,76.86354,346.56,0.58,0.7,2026-06-02,9,N20,VIIRS,n,2.0HRT,285.02,7.19,D

65.82961,76.86593,342.59,0.58,0.7,2026-06-02,9,N20,VIIRS,n,2.0HRT,284.39,7.19,D

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66.25023,76.82342,346.1,0.55,0.68,2026-06-02,9,N20,VIIRS,n,2.0HRT,284.12,5.18,D

66.34604,77.53783,327.09,0.55,0.68,2026-06-02,9,N20,VIIRS,n,2.0HRT,281.6,1.92,D

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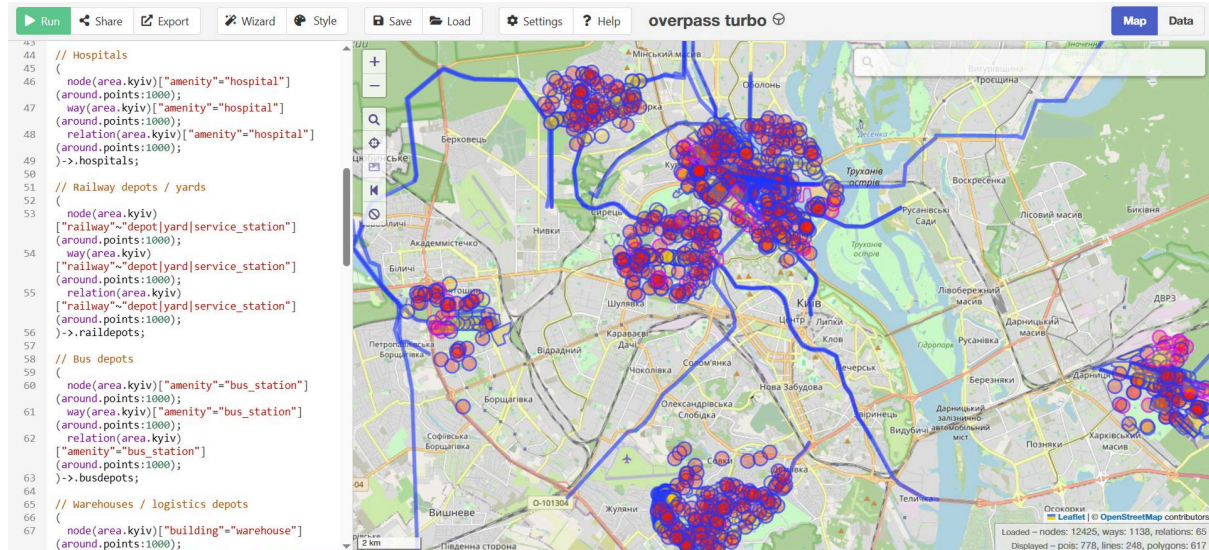
70.38127,68.40747,338.38,0.48,0.64,2026-06-02,9,N20,VIIRS,n,2.0HRT,277.15,3.25,D

<https://firms.modaps.eosdis.nasa.gov/api/>

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	fid	latitude	longitude	bright_ti4	scan	track	acq_date	acq_time	satellite	instrument	confidence	version	bright_ti5	frp	daynight
2	79	50.46605	30.46447	299.75	0.4	0.37	#####	18 N20	VIIRS	n	2.0NRT	282.93	0.79	N	
3	78	50.42737	30.67949	328.48	0.4	0.37	#####	18 N20	VIIRS	n	2.0NRT	283.8	3.44	N	
4	74	50.41422	30.35901	311.03	0.4	0.37	#####	18 N20	VIIRS	n	2.0NRT	280.81	0.95	N	
5	30105	50.5083	30.42545	308.62	0.5	0.49	#####	2326 N21	VIIRS	n	2.0NRT	282.69	1.7	N	
6	73	50.40009	30.4769	326.99	0.4	0.37	#####	18 N20	VIIRS	n	2.0NRT	281.35	3.55	N	
7	30104	50.48942	30.5075	309.33	0.5	0.49	#####	2326 N21	VIIRS	n	2.0NRT	283.56	1.17	N	
8	30103	50.4879	30.50781	320.63	0.5	0.49	#####	2326 N21	VIIRS	n	2.0NRT	283.71	3.07	N	
9	30102	50.48626	30.48561	331.08	0.5	0.49	#####	2326 N21	VIIRS	n	2.0NRT	284.28	5.43	N	
10	143	50.50789	30.42643	298.5	0.37	0.58	#####	107 N21	VIIRS	n	2.0NRT	279.83	1.96	N	
11	30101	50.48492	30.50677	308.61	0.5	0.49	#####	2326 N21	VIIRS	n	2.0NRT	282.91	1.17	N	
12	30100	50.48484	30.48599	332.01	0.5	0.49	#####	2326 N21	VIIRS	n	2.0NRT	283.68	4.2	N	
13	30099	50.48341	30.50709	295.22	0.5	0.49	#####	2326 N21	VIIRS	n	2.0NRT	282.9	0.48	N	
14	134	50.39227	30.47821	297.92	0.37	0.58	#####	107 N21	VIIRS	n	2.0NRT	281.61	0.74	N	
15	133	50.39187	30.47721	299.48	0.37	0.58	#####	107 N21	VIIRS	n	2.0NRT	281.09	0.94	N	
16	132	50.38289	30.33216	320.92	0.37	0.58	#####	107 N21	VIIRS	n	2.0NRT	278.19	2.28	N	
17	30098	50.47502	30.51944	337.13	0.5	0.49	#####	2326 N21	VIIRS	n	2.0NRT	288.72	4.34	N	
18	30097	50.47345	30.51977	336.7	0.5	0.49	#####	2326 N21	VIIRS	n	2.0NRT	287.72	7.5	N	
19	142	50.50647	30.43118	316.9	0.37	0.58	#####	107 N21	VIIRS	n	2.0NRT	282.04	2.49	N	
20	30096	50.46367	30.46224	304.52	0.5	0.49	#####	2326 N21	VIIRS	n	2.0NRT	283.55	1.48	N	
21	141	50.4744	30.51832	324.65	0.37	0.58	#####	107 N21	VIIRS	n	2.0NRT	282.75	2.41	N	
22	140	50.46211	30.46081	314.02	0.37	0.58	#####	107 N21	VIIRS	n	2.0NRT	281.77	1.69	N	
23	139	50.4435	30.36598	300.17	0.36	0.58	#####	107 N21	VIIRS	n	2.0NRT	279.56	0.51	N	
24	138	50.42686	30.67582	318.1	0.38	0.58	#####	107 N21	VIIRS	n	2.0NRT	280.04	3.14	N	
25	137	50.42541	30.68063	326.67	0.38	0.58	#####	107 N21	VIIRS	n	2.0NRT	279.82	3.14	N	
26	136	50.4	30.48953	313.99	0.37	0.58	#####	107 N21	VIIRS	n	2.0NRT	280.46	1.81	N	
27	135	50.39858	30.49419	309.3	0.37	0.58	#####	107 N21	VIIRS	n	2.0NRT	280.98	1.81	N	
28	30095	50.47345	30.51977	336.7	0.5	0.49	#####	2326 N21	VIIRS	n	2.0NRT	287.72	7.5	N	
Kyiv Hot Spots 01 02 June															

2. Spatial Analysis

I extracted the Kyiv restricted infrastructure data by querying OpenStreetMap through Overpass Turbo. I set the search for infrastructure within 375 m (primary zone) and 1000 m (secondary zone) separately.



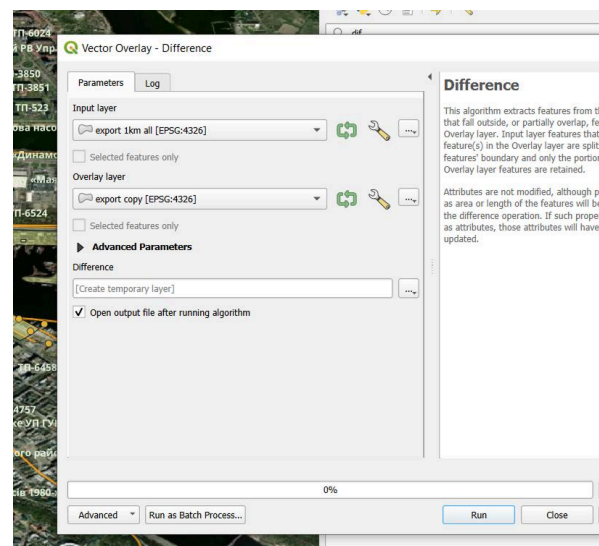
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2	relation/1	relation/1966480															
3	relation/1	relation/1993434						hospital									
4	relation/4	relation/4268400						hospital									
5	relation/4	relation/4851166															
6	relation/4	relation/4855122															
7	relation/5	relation/5328413						hospital									
8	relation/6	relation/6366872						police									
9	relation/6	relation/6366873															
10	relation/6	relation/6366875			19	3115	Д²ÑfD»D,Ñ+Ñ D»DµD' D%Ñ»D* D\$Ñ»D,Ñ+DµD²ÑÑ»CED»D%D³D%										
11	relation/6	relation/6366879															
12	relation/6	relation/6620904			1		Ð—D²D³D²	hospital									
13	relation/6	relation/6679675															
14	relation/6	relation/6879331			8	4050	Д²ÑfD»D,Ñ+Ñ D»DµD' D%Ñ»D* D\$Ñ»D,Ñ+DµD²ÑÑ»CED»D%D³D%	hospital									
15	relation/8	relation/8524662															
16	relation/9	relation/9300922			3		Д²ÑfD»D,Ñ+Ñ D»DµD' D%Ñ»D* D\$Ñ»D,Ñ+DµD²ÑÑ»CED»D%D³D%										
17	relation/1	relation/16354461															
18	relation/1	relation/16366708															
19	relation/1	relation/16909030															
20	relation/1	relation/16909031															
21	relation/1	relation/16909032															
22	relation/1	relation/16909033															
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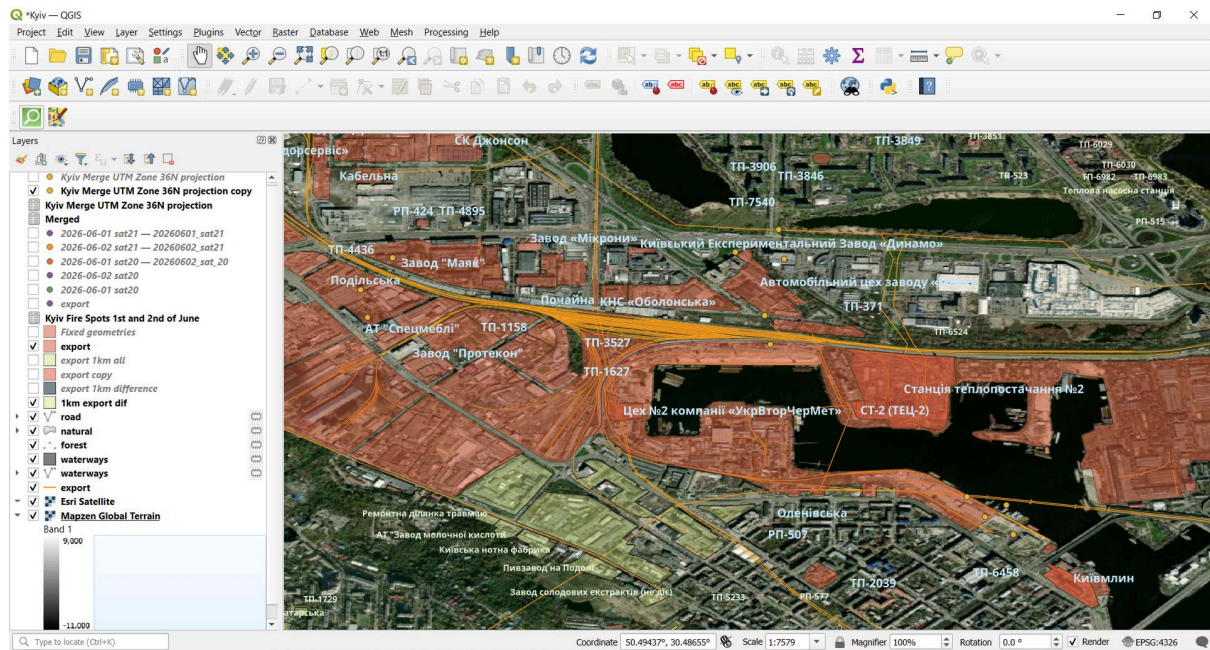
3. Infrastructure extraction

I exported the merged fire hotspots and the OSM data into QGIS. To differentiate the secondary-zone I create a difference layer between the 1000 m results minus 375 m results.

I then removed duplicate infrastructure already present in the primary zone.

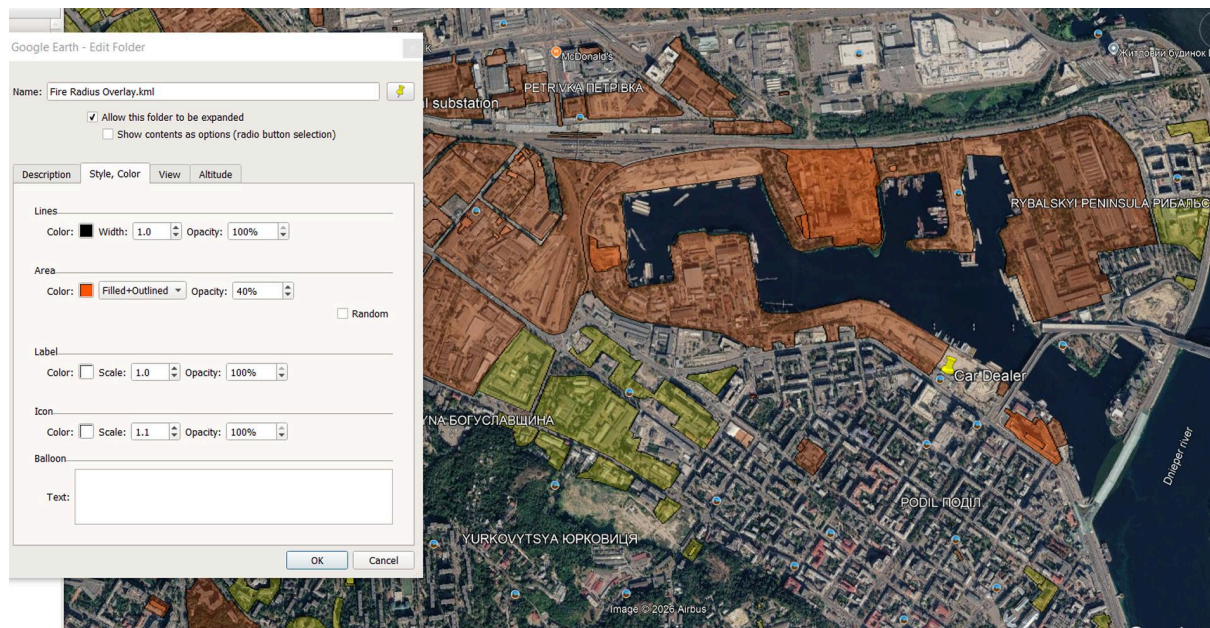
I changed the styles to distinguish each layer.





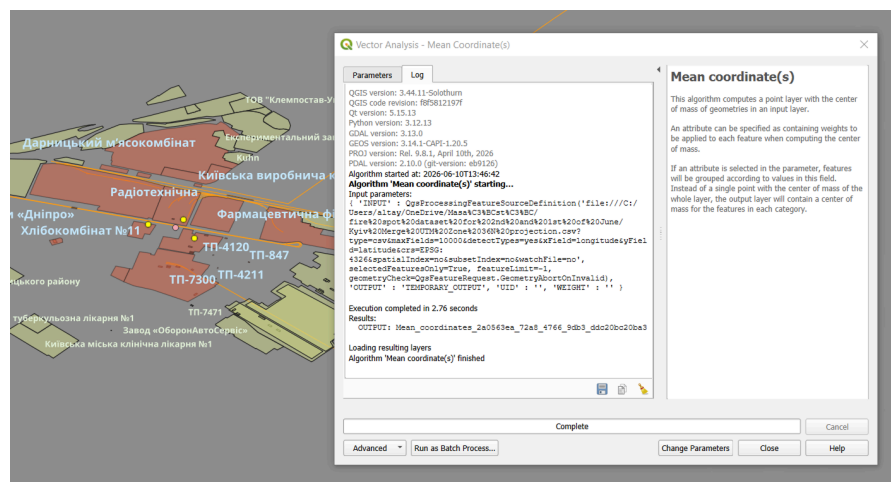
4. Output Generation

I created the Geospatial deliverables by exporting the infrastructure layers as KML. I then stylised them in Google Earth Pro and saved the overlays once again.



5. Cluster Averaging

I manually selected and averaged the mean coordinates of clusters.



6. Database Construction

I created the infrastructure database by first converting polygons and multipolygons to centroid coordinates, and then converting GeoJSON outputs into structured CSV tables using AI. I extracted the name, infrastructure type, coordinates and the operator where available. I deleted all the nameless locations and insignificant nodes to lean out the dataset.

infrastructure_centroids.csv

showing first 200 of 233 populated rows.

#	A	B	C	D	E	F
1	name	type	subtype	operator	lat	lon
2	Національний інститут раку	hospital	amenity=hospital	МОЗ України	50.38878067883904	30.48483540945952
3	Інститут отоларингології	hospital	amenity=hospital		50.455560853106775	30.459539706204787
4	Автобусний парк №8	industrial	landuse=industrial	Київпастранс	50.46559029989716	30.45881239698154
5	Київська міська клінічна лікарня №14	hospital	amenity=hospital		50.456887079406954	30.458262983067588
6	Кінологічний центр МВС в метрополітені	police	amenity=police	МВС України	50.446108245914175	30.363901938108263
7	Роза-Хутро	industrial	landuse=industrial		50.4421654301741	30.363047366703835
8	Територія колишнього святошинського молочного заводу	industrial	landuse=industrial		50.4494659931762	30.366667669478826
9	Київська обласна клінічна лікарня №1	hospital	amenity=hospital		50.4729286785747	30.468175799896162
10	Радіотехнічна	substation	power=substation	Київські електромережі	50.43097679316797	30.68100404921832
11	Всеукраїнський центр материнства та дитинства	hospital	amenity=hospital	НАМН України	50.468778794697954	30.474967272265808
12	Подільська	substation	power=substation	Київські електромережі	50.4860690024141	30.48588712137252
13	Дарницький міжкомбінат	industrial	landuse=industrial		50.434699706411116	30.67851520539961
14	Завод "Протекон"	industrial			50.482986335361076	30.494137562267206
15	Пікетир Сервіс	industrial			50.48371964445807	30.494245433719392

7. Classification

In order to filter through the most important data, I created a “priority” list to act as a ranking column. The vital institutional buildings were classified as “Strategic” targets for the attacking military while other key infrastructure was classified as “Operational”. Everything that fell outside of both were classified as “Background”. This was created in two excel sheets, one for the primary and the other for the secondary radius.

infrastructure_damage_assessment_Kyiv0...

Strategic

type

subtype

=IF(
OR(
ISNUMBER(SEARCH("hospital",D2)),
ISNUMBER(SEARCH("depot",D2)),
ISNUMBER(SEARCH("yard",D2)),
ISNUMBER(SEARCH("power",D2)),
ISNUMBER(SEARCH("substation",D2)),
ISNUMBER(SEARCH("plant",D2)),
ISNUMBER(SEARCH("fire_station",D2)),
ISNUMBER(SEARCH("police",D2))
),
"Strategic",
IF(
OR(
ISNUMBER(SEARCH("industrial",D2)),
ISNUMBER(SEARCH("warehouse",D2))
),
"Background",
"Operational"
)
)

+ Add new function

Ctrl + Alt + N

Strategic

police

amenity=

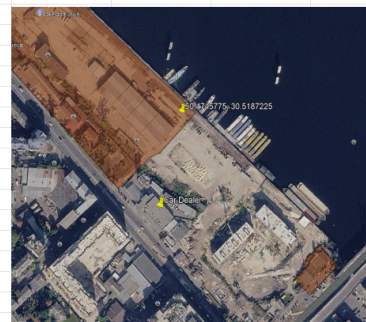
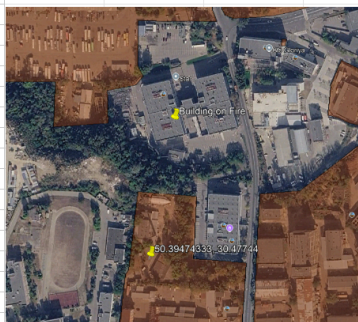
#	A	B	C	D	E	F	G	H
1	name	translated name	priority	type	subtype	operator	lat	lon
2	Національний інститут раку	National Cancer Institute	Strategic	hospital	amenity=hospital	МОЗ України	50.38878068	30.48483541
3	Інститут отоларингології	Institute of Otolaryngology	Strategic	hospital	amenity=hospital		50.45556085	30.45953971
4	Київська міська клінічна лікарня №14	Kyiv City Clinical Hospital No. 14	Strategic	hospital	amenity=hospital		50.45688708	30.45826298
5	Київська обласна клінічна лікарня №1	Kyiv Regional Clinical Hospital No. 1	Strategic	hospital	amenity=hospital		50.47292868	30.4681758
6	Всеукраїнський центр материнства та дитинства	All-Ukrainian Center for Motherhood and Childhood	Strategic	hospital	amenity=hospital	НАМН України	50.46877879	30.47496727
7	Університетська клініка КНУ	University Clinic of KNU	Strategic	hospital	amenity=hospital		50.38925305	30.474392
8	Київська міська дитяча клінічна інфекційна лікарня	Kyiv City Children's Clinical Infectious Diseases Hospital	Strategic	hospital	amenity=hospital		50.48234271	30.45743787
9	Центр інфекційних уражень нервової системи	Center for Infectious Diseases of the Nervous System	Strategic	hospital	amenity=hospital	Інститут епідеміології та інфекційних	50.510675	30.43994347
10	Територіальне медичне об'єднання "Фізіотерпія"	Territorial Medical Association "Phisiology"	Strategic	hospital	amenity=hospital		50.3918614	30.48704735
11	Дитяча клінічна лікарня №8	Children's Clinical Hospital No. 8	Strategic	hospital	amenity=hospital		50.46616466	30.4756655
12	Святошинський психо-неврологічний інтернат для жінок	Svyatoshyn Psychoneurological Boarding School for Women	Strategic	hospital	amenity=hospital		50.44988348	30.37014555
13	Оберіг	Amulet	Strategic	hospital	amenity=hospital		50.45693823	30.45815675
14	Клініка Варта	Clinic Warta	Strategic	hospital	amenity=hospital		50.3896242	30.4832194
15	Психолог Олександр Шумський	Psychologist Olesky Shumsky	Strategic	hospital	amenity=hospital		50.4684067	30.5132534
16	Оберіг	Amulet	Strategic	hospital	amenity=hospital		50.4685159	30.4582577
17	Станція теплопостачання №2	Heat supply station No. 2	Strategic	plant	power=plant		50.481147	30.51404907
18	Кінологічний центр МВС в метрополітені	The Ministry of Internal Affairs' dog training center in the metro	Strategic	police	amenity=police	МВС України	50.44610825	30.36390194
19	Подільське УП ГУНП	Podilskiy UP Main Police Department	Strategic	police	amenity=police		50.4696644	30.51718273
20	Шевченківське УП ГУНП	Shevchenkivske UP Main Police Department	Strategic	police	amenity=police		50.46952187	30.47176911
21	Відділення міліції Подільського району	Podilskiy district police department	Strategic	police	amenity=police		50.5128321	30.42810196
22	Управління по боротьбі з кіберзлочинністю ГУ МВС України в м. Києві	Department for Combating Cybercrime of the Main Directorate of the Ministry of Internal Affairs in Kyiv	Strategic	police	amenity=police		50.471913	30.46324995
23	Відділ по збіву зброї	Weapons Trafficking Department	Strategic	police	amenity=police		50.5133994	30.4299409
24	УПО в м. Києві Шевченківського та Подільського РВ	UPR in Kyiv Shevchenkiyskiy and Podilskiy Autonomous Okrugs	Strategic	police	amenity=police		50.4696662	30.5140316
25	Територіальні сервісні центри МВС 8041	Territorial service centers of the Ministry of Internal Affairs 8041	Strategic	police	amenity=police		50.4423026	30.3671426
26	Державна пожежна частина №7 Подільського району	State Fire Department No. 7 of Podilskiy District	Strategic	fire_station	amenity=fire_station	ДСНС України у м. Києві	50.50779757	30.42478433
27	Фармацевтична	Pharmaceutical	Operational	halt	railway=halt	Південно-Західна залізниця	50.4297055	30.6854241
28	Курєнівка	Kurenivka	Operational	halt	railway=halt	Українізція	50.4894195	30.4827922
29	ПС Північна - 330 - Ірпінь, ПС Біличі	Pivnichna PS - 330 - Irpin PS, Bilichy PS	Operational	line	power=line	Київобленерго	50.56383608	30.4538707
30	ПС Новокиївська-330 - ПС Нивки-330	PS Novokivska-330 - PS Nivky-330	Operational	line	power=line	Київобленерго	50.43154506	30.368354

8. Verification Via Cluster Analysis

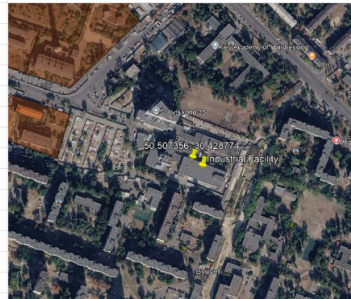
Seven independently geolocated damage locations were compared against FIRMS-derived cluster centroids. Distances ranged from 0–150 m, with an average offset of approximately 73 m, which is remarkably tight. A single VIIRS detection can have hundreds of metres of uncertainty. However, averaging multiple points seems to reduce signal noise.

FIRMS cluster centroids turned out to be very useful in approximating the centre of significant fire activity generated by a strike. This is encouraging, since a geolocator sent to a point 137 m from a confirmed fire would almost certainly find the correct site within minutes. Everything converged on the right neighbourhood, and often on the right block. For a method built from a simple approach, that is a surprisingly strong result by itself. The larger distance findings which ranged 137–150 m were still in the damage areas.

SOUTHER CLUSTER 1: OFFICE BUILDING			SOUTHER CLUSTER 2: POSTAL SERVICE			NORTHERN CLUSTER 1: PORTSIDE CAR DEALER		
latitude	longitude		latitude	longitude		latitude	longitude	
50.40009	30.4769		50.39858	30.49419		50.47544	30.51736	
50.39187	30.47721		50.4	30.48953		50.47345	30.51977	
50.39227	30.47821		average			50.47502	30.51944	
average			50.39929	30.49186		50.4744	30.51832	
50.39474333	30.47744		real location			average		
50.39666732	30.47814237		50.398958	30.491092		50.4745775	30.5187225	
distance			distance			real location		
150 meters far			60 meters far (almost exact)			50.47326	30.51825	
source			source			137 meter far		
https://x.com/MESmaxie/status/2061719641060184424			CIR: Entry Number UW82925			source		
			https://www.info-res.org/eyes-on-russia/maps/eyes-on-russia-map/			https://x.com/zarGEOINT/status/2061649998761128026		



EASTERN CLUSTER: GAS STATION			NORTH WEST CLUSTER: INDUSTRIAL SITE			NORTH CLUSTER 2: ELECTRICAL SUBSTATION		
latitude	longitude		latitude	longitude		latitude	longitude	
50.47544	30.51736		50.50542	30.42968		50.48796	30.4873	
50.47345	30.51977		50.5087	30.43113		50.48484	30.48599	
50.47502	30.51944		50.5083	30.42545		50.48626	30.48561	
50.4744	30.51832		50.50647	30.43118		average		
average			50.50789	30.42643		50.48635333	30.4863	
50.4745775	30.5187225		average			real location		
real location			50.507356	30.428774		50.486796	30.485736	
50.47326	30.51825		real location			60 meters far (almost exact)		
40 meters far (practically exact)			50.507194	30.429006		source		
source			exact			CIR: Entry Number UW83042		
https://x.com/stopthetwigs/status/2061611104158015866			source			https://www.info-res.org/eyes-on-russia/maps/eyes-on-russia-map/		
			CIR: Entry Number UW82938					
			https://www.info-res.org/eyes-on-russia/maps/eyes-on-russia-map/					



From: [Cluster Study.xlsx](#)

CENTRAL KYIV CLUSTER: DEFENCE MANUFACTURER	
latitude	longitude
50.46605	30.46447
50.46367	30.46224
50.46211	30.46081
average	
50.46394333	30.46250667
real location	
50.4633173	30.4631062
distance	
65 meters far (almost exact)	
source	
https://x.com/AMK_Mapping/status/2061658415605452866	



Cases	Av. vs Confirmed Distance
Industrial Site	0 m
Gas Station	40 m
Electrical Substation	60 m
Postal Service	60 m
Defence Manufacturer	65 m
Car Dealership	137 m
Office Building	150 m

Case Study

The hotspots correspond to areas where damage was reported. An example is Clinic No. 7. Although the facility fell outside the primary search radius, it was located near a hotspot cluster and was captured by the secondary review zone. This illustrates how the dataset can identify and highlight relevant areas warranting further investigation even when individual facilities fall outside the nominal detection radius.

Source: https://mezha.net/eng/bukvy/946d24e1_mass_attack_damaged/



Public reporting identified several other healthcare facilities damaged during the attack. The hotspot-derived search areas overlap with the districts where those facilities were reported:

General Practice / Family Medicine Clinic No. 5 (Holosiivskyi district): reported as heavily damaged, with upper floors destroyed.

Ambulatory No. 10 and No. 11(Podilskyi district): windows and rooms damaged.

Center for Primary Health Care No. 3 (Sviatoshynskyi district): building damaged.

Given that the highlights correspond to areas where damage was independently reported:

- The methodology identifies infrastructure in bombed areas.
- The methodology can surface leads that investigators can verify further.

Even a small cluster proved to be effective in aiding geolocation, like the City Clinic Hospital.

Source: <https://en.interfax.com.ua/news/general/1172907.html>



Breakdown of Major Affected Zones:

- 1. Central Kyiv belt (Shevchenkivskiy–Podil–Pechersk interface)
- 2. Eastern residential belt (Darnytskyi)
- 3. Western residential edge (Sviatoshyn / Holosiiv fringe)
- 4. Northern industrial corridor (Obolon)

Breakdown of Kyiv Area Fire Clusters

Reported	Hotspot pattern
Holosiivskiy damage	Western and south-central clusters
Podilskiy damage	Central-right-bank clusters
Sviatoshynskiy damage	Western cluster
Industrial fires	Northern industrial corridor
Medical facility damage	Multiple hospitals and clinics identified

The methodology identified different damage environments:

Eastern cluster:

- Residential areas.
- Apartment districts.
- Urban infrastructure.

This suggests civilian-area effects.

Northern cluster:

- Warehouses.
- Industrial land.
- Transport infrastructure.

This suggests industrial and logistics zones.

General Notes:

- Attacks occurred across both residential and industrial environments.
- The northern corridor contained a disproportionately high concentration of logistics and industrial infrastructure while the eastern belt was dominated by civilian residences.
- Healthcare facilities appeared repeatedly across multiple clusters.

Conclusion

The methodology successfully reduced a large urban battlespace into a manageable set of investigation priorities and demonstrated promising correspondence with independently verified damage locations. FIRMS cluster centroid analysis has shown to be a useful rapid geolocation aid during Mass Attacks into urban environments.

Discussion:

There are many more ways to improve upon this model. For starters, only the OSM infrastructure database was used in this study. More datasets must be combined to form a larger pool of infrastructure. The most important additions would be the strategic locations, which are the government and especially the military zones, which were not included in this study. Supply chain maps should be overlaid with OSM if possible to contextualise strikes on operational routes.

Secondly, mass attacks can also be detected through other means than VIIRS, which has many limitations. Some sites which were heavily bombed did not register on the FIRMS website, which shows a blind spot for the current model. [Ollie Ballinger's proxy map](#) might be another avenue to look at, which is a tool that looks at damage probabilities on a map based on the rate of change in Sentinel-1 collected imagery within a given country.

If these two parts of the methodology are improved and expanded upon, it will have the potential to yield a clear and precise picture of an entire bombing campaign. Although specific verification will always be necessary, Massive Attack Mapping Methodology has the potential to contribute to this field.